

Answers

Q1

To analyze the temperature change on May 20th, we note that the temperature at morning ($t=0$) is 19°C and at afternoon ($t=12$) is 23°C . Modeling this, the interval $[0, 12]$ follows the linear growth trend

$$T(t) = \frac{t^2 - 144}{t - 12} + 31$$

and the interval $[12, 24]$ follows the atmospheric cooling cycle

$$T(t) = k \cos\left(\frac{\pi t}{12}\right) + 25.$$

Derive the equation to ensure the temperature function $T(t)$ is continuous at the boundary point $t = 12$ and find the value of the constant k .

$$\lim_{t \rightarrow 12^-} T(t) = T(12)$$

$$t^2 - 144 = (t - 12)(t + 12)$$

$$\frac{t^2 - 144}{t - 12} = t + 12$$

$$T(t) = t + 12 + 31$$

$$= t + 43$$

$$\lim_{t \rightarrow 12^-} T(t) = 55$$

$$T(12) = k \cos(\pi) + 25$$

$$= -k + 25$$

$$55 = -k + 25$$

$$k = -30$$

Q2

From May 22 to 25, afternoon temperatures show a linear increase of 1°C per day. Based on this short-term trend and discrete data points, researchers derived the prediction function

$$T_{\text{after}}(t) = \frac{75t^2 + 150t + 22}{3t^2 + 1}.$$

Calculate the base temperature (limit value) that the afternoon temperature will converge to as time progresses infinitely ($t \rightarrow \infty$).

$$\lim_{t \rightarrow \infty} \frac{75t^2 + 150t + 22}{3t^2 + 1} = \frac{75}{3}$$

$$= 25^\circ\text{C}$$

Q3

Actual afternoon temperatures for May 18 ($t=0$), 19 ($t=1$), and 20 ($t=2$) are 27°C , 25°C , and 23°C respectively.

(a) Find the average rate of change of temperature from $t = 0$ to $t = 2$.

(b) Given the polynomial

$$T(t) = 27 - 2t$$

that passes through these points, use the definition of the derivative to find the instantaneous rate of change at $t = 1$ (May 19).

$$a) \frac{23 - 27}{2 - 0} = -2$$

$$b) T'(t) = \lim_{h \rightarrow 0} \frac{T(t+h) - T(t)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{27 - 2(t+h) - (27 - 2t)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-2h}{h} = -2$$

Q4

Using May 20 precipitation data, the probability of rain is observed to be:

- 30% at 6 AM ($t = 6$)
- 60% at 12 PM ($t = 12$)

For $0 \leq t \leq 24$, the morning precipitation model ($t \leq 12$) is defined by

$$P(t) = a(t - 12)^2 + 30(t - 12) + b$$

and the afternoon model ($t > 12$) is defined by

$$P(t) = 30 \ln(t - 11) + 60.$$

Find the values of constants a and b , and verify whether $P(t)$ is differentiable at $t = 12$.

$$P(6) = 30$$

$$30 = 36a - 120$$

$$P(12) = 60$$

$$a = \frac{25}{6}$$

$$60 = a(0)^2 + 30(0) + b$$

$$b = 60$$

$$30 = a(6-12)^2 + 30(6-12) + 60$$

Q5

On May 18, the afternoon temperature is 27°C , with a rate of change

$$T'(t) = 1.5^\circ\text{C}/\text{hour}.$$

Data indicates power consumption $E(T)$ where

$$E(27) = 1200 \text{ MW}$$

and

$$\frac{dE}{dT} = 40 \text{ MW}/^\circ\text{C}$$

at $T = 27$. Using the Chain Rule, derive and calculate the rate of change of power consumption per hour (dE/dt) at this moment.

$$\frac{dE}{dt} = \frac{dE}{dT} \times \frac{dT}{dt}$$

$$= 40(1.5)$$

$$= 60$$

$$60 \text{ MW}/\text{hour}$$

Q6

Modeling May 21st data ($x = 20$ morning temp, $y = 60$ afternoon rain probability) results in the implicit relation

$$x^3 - xy + y^2 = 10400.$$

Differentiate both sides with respect to x to derive the expression for dy/dx , and find the instantaneous rate of change at the data point $(20, 60)$.

$$3x^2 - (x \frac{dy}{dx} + y) + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{y - 3x^2}{-x + 2y}$$

$$\frac{60 - 1200}{-20 + 120} = -11.4$$

Q7

A cylindrical reservoir (radius $r = 10 \text{ m}$ fixed) is filling with rain. Sensors measure water level h rising at

$$\frac{dh}{dt} = 0.05 \text{ m}/\text{hour}.$$

From the cylinder volume formula

$$V = \pi r^2 h,$$

derive the related rates equation for time t and calculate the rate of change of volume dV/dt .

$$\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$$

$$= \pi (10)^2 (0.05) = 5\pi$$

Answers

Q8

On May 24th, morning ($t = 0$) temperature is 17°C with an initial rise rate

$$T'(0) = 1.2^\circ\text{C}/\text{hour}.$$

Using these data points, derive the tangent line equation (linear approximation function) $L(t)$ at $t = 0$, and predict the temperature 30 minutes later ($t = 0.5$ hours).

$$L(t) = 17 + 1.2t$$

$$L(0.5) = 17.6$$

Q9

Peak power load data (MW) for May 15-17 are 753.33, 728.60, and 735.92. The continuous load model is

$$K(t) = 16t^3 - 36t^2 + 753.33 \quad (0 \leq t \leq 2).$$

Find the critical point using $K'(t)$ and determine the time t when the peak load is at a local minimum and the corresponding minimum peak power value.

$$K'(t) = 48t^2 - 72t$$

$$48(t - 1.5) = 0$$

$$t = 1.5$$

$$K''(1.5) > 0$$

$$K(1.5) = 726.33$$

Q10

Power supply (MW) for May 12-15:

1082.99 $\rightarrow$ 1142.01 $\rightarrow$ 1153.93 $\rightarrow$ 1070.27.

Continuous function:

$$S(t) = -9t^3 + 27t^2 + 41t + 1082.99 \quad (0 \leq t \leq 3).$$

Derive the second derivative $S''(t)$ to find the inflection point where the supply increase rate begins to decrease.

$$S'(t) = -27t^2 + 54t + 41$$

$$S''(t) = -54t + 54$$

$$-54t + 54 = 0$$

$$t = 1$$

Q11

Using the Jeju Weather data, the afternoon temperatures for May 18-20 are recorded as:

$$T(0) = 27^\circ\text{C}, \quad T(1) = 25^\circ\text{C}, \quad T(2) = 23^\circ\text{C}$$

where t represents days after May 18.

(a) Using the finite difference definition

$$R(t) = \frac{T(t+1) - T(t)}{1},$$

calculate the daily rate of temperature change $R(0)$ and $R(1)$.

(b) Using these rates, construct the Left Riemann Sum approximation for

$$\int_0^2 R(t) dt$$

and determine whether the accumulated change matches the actual overall temperature displacement from May 18 to May 20.

$$a) R(0) = \frac{25 - 27}{1 - 0} = -2$$

$$R(1) = \frac{23 - 25}{2 - 1} = -2$$

b)

Left Riemann Sum:

$$R(0)(1) + R(1)(1)$$

$$= (-2)(1) + (-2)(1)$$

$$= -4$$

Total displacement:

$$23 - 27 = -4$$

Matches

Q12

Reserve data: 316.94 MW (May 15), 394.01 MW (May 17). Derived rate:

$$R'(t) = -6t^2 + 24t + 29.5.$$

(a) Apply the Fundamental Theorem of Calculus (FTC) to derive the antiderivative of the integral

$$\int_0^2 R'(t) dt.$$

(b) Perform the algebraic calculation and find the absolute error compared to the actual net change of 77.07 MW.

$$R(t) = -2t^3 + 12t^2 + 29.5t + C$$

$$\int_0^2 R'(t) dt = [-2t^3 + 12t^2 + 29.5t]_0^2$$

$$= -16 + 48 + 59$$

$$= 91$$

$$|91 - 77.07| = 13.93$$

Q13

Supply growth follows

$$\frac{dS}{dt} = k(2162.81 - S).$$

Given the initial condition

$$S(0) = 1090.94$$

and the observed data

$$S(1) = 1129.93,$$

use separation of variables to derive the prediction model and find the exact value of the constant k .

$$\frac{dS}{2162.81 - S} = k dt$$

$$-\ln|2162.81 - S| = kt + C$$

$$S(t) = 2162.81 - Ce^{-kt}$$

$$1090.94 = 2162.81 - C$$

$$C = 1071.87$$

$$S(t) = 2162.81 - 1071.87e^{-kt}$$

$$S(1) = 1129.93 = 2162.81 - 1071.87e^{-k(1)}$$

$$e^{-k} = \frac{1032.88}{1071.87}$$

$$k = -\ln\left(\frac{1032.88}{1071.87}\right)$$

Answers

Q14

$$\frac{dy}{dt} = \frac{t^2 - 121}{y}$$

(a) Use implicit differentiation to derive the second derivative

$$\frac{d^2y}{dt^2}$$

(b) Calculate the value of

$$\frac{d^2y}{dt^2}$$

at the point (11, 378.60) from May 11th data and determine the concavity of the solution curve at that point.

$$\frac{d^2y}{dt^2} = \frac{2ty - (t^2 - 121) \frac{dy}{dt}}{y^2}$$

$$(11)^2 - 121 = 0$$

$$\frac{d^2y}{dt^2} = \frac{2(11)(378.60)}{(378.60)^2} > 0$$

Concave up

Q15

Path A and Path B are modeled by the functions

$$f(x) = -x^2 + 4x + 35$$

and

$$g(x) = x^2 - 4x + 23.$$

Set up and evaluate the definite integral representing the area enclosed between the two curves. Round your answer to three decimal places.

$$-x^2 + 4x + 35 = x^2 - 4x + 23$$

$$2x^2 - 8x - 12 = 0$$

$$x^2 - 4x - 6 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 24}}{2}$$

$$= \frac{4 \pm 2\sqrt{10}}{2}$$

$$= 2 \pm \sqrt{10}$$

$$\text{Area} = \frac{1}{2} \int_{2-\sqrt{10}}^{2+\sqrt{10}} [f(x) - g(x)] dx$$

$$= \frac{1}{2} \int_{2-\sqrt{10}}^{2+\sqrt{10}} (-2x^2 + 8x + 12) dx$$

$$= 42.163$$

Q16

Coastline radar boundary:

$$y = \sqrt{x} \quad (0 \leq x \leq 4)$$

Derive the Disk Method integral formula to calculate the volume of the solid formed by rotating this region around the x-axis, and calculate the final volume.

$$\pi \int_0^4 (\sqrt{x})^2 dx$$

$$= \pi \left[\frac{x^2}{2} \right]_0^4$$

$$= \pi \left(\frac{16}{2} \right)$$

$$= 8\pi$$

Q17

Drone coordinates:

$$x(t) = 3t^2, \quad y(t) = 2t^3 \quad (0 \leq t \leq 2)$$

Derive the arc length formula structure, substitute the derivatives to obtain a u-substitution integral, and perform the algebraic calculation to find the total distance traveled by the drone over 2 hours.

$$\int_0^2 \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

$$x'(t) = 6t$$

$$y'(t) = 6t^2$$

$$\int_0^2 \sqrt{(6t)^2 + (6t^2)^2} dt$$

$$= \int_0^2 6t \sqrt{1+t^2} dt$$

$$u = 1+t^2$$

$$du = 2t dt$$

$$\int_1^5 3\sqrt{u} du$$

$$= 20.360$$

Q18

Echo diffusion boundary:

$$r = 2\sqrt{\theta}$$

Substitute this model into the polar area formula, derive the definite integral, and calculate the area of the cloud echo region for

$$0 \leq \theta \leq \pi.$$

$$\frac{1}{2} \int_0^\pi (2\sqrt{\theta})^2 d\theta$$

$$= 2 \int_0^\pi \theta d\theta$$

$$= 2 \left[\frac{\theta^2}{2} \right]_0^\pi$$

$$= \pi^2 \text{ or } 9.87$$

Q19

Data:

$$T(0) = 27, \quad T(1) = 25, \quad T(2) = 23, \quad T(3) = 22.$$

Initial derivative values at $t = 0$ derived as:

$$T(0) = 27, \quad T'(0) = -2, \quad T''(0) = 0, \quad T'''(0) = 1.$$

Derive the 3rd-degree Taylor polynomial

$$P_3(t)$$

centered at $t = 0$, and find the absolute error between the predicted value at $t = 2$ and the actual data (23°C).

$$P_3(t) = T(0) + T'(0)t + \frac{T''(0)t^2}{2!} + \frac{T'''(0)t^3}{3!}$$

$$P_3(t) = 27 - 2t + 0 + \frac{1}{6}t^3$$

$$P_3(2) = 27 - 4 + \frac{8}{6} = 24.33$$

$$|24.33 - 23| = 1.33$$

Answers

Q 20

Data:

$$S(0) = 1153.93, \quad S(1) = 1070.27, \quad S(2) = 1090.94.$$

Derived values at $t = 0$:

$$S(0) = 1153.93, \quad S'(0) = -83.66, \quad S''(0) = 104.33.$$

Use $P_2(t)$ to predict supply 3 days later ($t = 3$) and calculate the absolute error against the actual data (1129.93 MW).

$$P_2(t) = S(0) + S'(0)t + \frac{S''(0)t^2}{2!}$$

$$P_2(t) = 1153.93 - 83.66t + \frac{104.33}{2}t^2$$

$$\begin{aligned} P_2(3) &= 1153.93 - 83.66(3) + \frac{104.33}{2}(3)^2 \\ &= 1372.435 \end{aligned}$$

$$|1372.435 - 1129.93| = 242.505$$